



**KUALA LUMPUR UNIVERSITY
OF SCIENCE AND TECHNOLOGY**
(formerly known as Infrastructure University Kuala Lumpur)

ASSIGNMENT 1 : Network Topology & VLSM

Network Design Report

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1.0 Introduction

The goal of this assignment is to build a Local Area Network (LAN) for a company. The company has five departments: Production, Engineering, Sales, Finance, and HR.

The main tasks were:

1. Connect all departments using a Star Topology.
2. Use VLSM (Variable Length Subnet Masking) to create IP addresses.
3. Make sure all computers can "ping" each other.

I used the **Cisco Packet Tracer** software to design and test this network. The design uses one central router to manage all the traffic between the departments.

2.0 Hardware and Devices

I chose specific hardware to make sure the network works well.

2.1 Cisco Router (ISR 4331)

I used the Cisco 4331 Router. This is the "brain" of the network.

- **Why I chose it:** It is a strong router that lets us add extra parts (modules) if we need more ports.
- **The Problem:** The router only has three ports built-in.
- **The Solution:** I added a special card called the **NIM-ES2-4** module in Slot 2. This gave me four extra ports so I could connect all five departments.

2.2 Cisco Switches (2960)

Each department has its own Cisco 2960 Switch.

- **What it does:** It connects the PCs and Servers to the main Router.
- **Feature:** It supports VLANs, which I needed for the Sales, Finance, and HR departments.

2.3 Computers and Servers

Every department has one PC and one Server. I gave them static IP addresses so the numbers do not change.

3.0 VLSM Calculations (IP Addresses)

My Student ID ends in **03**, so my base network is **172.3.0.0**.

To save IP addresses, I used VLSM. This means I gave big departments more IPs and small departments fewer IPs. I sorted them from biggest to smallest.

Table 1: IP Addressing Plan (VLSM)

Department	Hosts Needed	Subnet Mask (CIDR)	Network Address	First Usable IP	Last Usable IP	Broadcast Address	Gateway IP
Production	3000	/20	172.3.0.0	172.3.0.1	172.3.15.254	172.3.15.255	172.3.0.1
Engineering	1000	/22	172.3.16.0	172.3.16.1	172.3.19.254	172.3.19.255	172.3.16.1
Sales	500	/23	172.3.20.0	172.3.20.1	172.3.21.254	172.3.21.255	172.3.20.1
Finance	100	/25	172.3.22.0	172.3.22.1	172.3.22.126	172.3.22.127	172.3.22.1
HR	50	/26	172.3.22.128	172.3.22.129	172.3.22.190	172.3.22.191	172.3.22.129

3.1 Detailed VLSM Calculation Process

To determine the correct subnet masks, I calculated the required host bits (n) using the formula $2^n - 2 \geq \text{Hosts}$.

A. Production Department (3000 Hosts)

- Requirement: 3000 usable IP addresses.
- Calculation:
 - $2^{11} = 2048$ (Too small)
 - $2^{12} = 4096$ (Sufficient)
- Host Bits (n): 12 bits are reserved for hosts.
- Network Bits: $32 - 12 = 20$.
- New CIDR: /20.
- Subnet Mask Binary: 11111111.11111111.11110000.00000000
- Subnet Mask Decimal: 255.255.240.0.
- Block Size: 16 (in the third octet).
- Range: 172.3.0.0 to 172.3.15.255.

B. Engineering Department (1000 Hosts)

- Requirement: 1000 usable IP addresses.
- Calculation:
 - $2^9 = 512$ (Too small)
 - $2^{10} = 1024$ (Sufficient)
- Host Bits (n): 10 bits.
- Network Bits: $32 - 10 = 22$.
- New CIDR: /22.
- Subnet Mask Binary: 11111111.11111111.11111100.00000000
- Subnet Mask Decimal: 255.255.252.0.
- Starting Network: 172.3.16.0 (The next available block).
- Range: 172.3.16.0 to 172.3.19.255.

C. Sales Department (500 Hosts)

- Requirement: 500 usable IP addresses.
- Calculation:
 - $2^8 = 256$ (Too small)
 - $2^9 = 512$ (Sufficient)
- Host Bits (n): 9 bits.
- Network Bits: $32 - 9 = 23$.
- New CIDR: /23.
- Subnet Mask Decimal: 255.255.254.0.
- Starting Network: 172.3.20.0.
- Range: 172.3.20.0 to 172.3.21.255.

D. Finance Department (100 Hosts)

- Requirement: 100 usable IP addresses.
- Calculation:
 - $2^6 = 64$ (Too small)
 - $2^7 = 128$ (Sufficient)
- Host Bits (n): 7 bits.
- Network Bits: $32 - 7 = 25$.
- New CIDR: /25.
- Subnet Mask Decimal: 255.255.255.128.
- Starting Network: 172.3.22.0.
- Range: 172.3.22.0 to 172.3.22.127.

E. HR Department (50 Hosts)

- Requirement: 50 usable IP addresses.
- Calculation:
 - $2^5 = 32$ (Too small)
 - $2^6 = 64$ (Sufficient)
- Host Bits (n): 6 bits.
- Network Bits: $32 - 6 = 26$.
- New CIDR: /26.
- Subnet Mask Decimal: 255.255.255.192.
- Starting Network: 172.3.22.128.
- Range: 172.3.22.128 to 172.3.22.191.

4.0 Router Configuration

This is the code I typed into the Router to make it work.

Plaintext

enable

configure terminal

hostname KL_Router

ip routing

! --- 1. Production Department ---

interface g0/0/0

ip address 172.3.0.1 255.255.240.0

no shutdown

exit

! --- 2. Engineering Department ---

```
interface g0/0/1

ip address 172.3.16.1 255.255.252.0

no shutdown

exit
```

! --- 3. Sales Department (VLAN 30) ---

```
interface vlan 30

ip address 172.3.20.1 255.255.254.0

no shutdown

exit

interface g0/2/0

switchport mode access

switchport access vlan 30

no shutdown

exit
```

! --- 4. Finance Department (VLAN 40) ---

```
interface vlan 40

ip address 172.3.22.1 255.255.255.128

no shutdown

exit

interface g0/2/1
```

```
switchport mode access  
  
switchport access vlan 40  
  
no shutdown  
  
exit
```

! --- 5. HR Department (VLAN 50) ---

```
interface vlan 50  
  
ip address 172.3.22.129 255.255.255.192  
  
no shutdown  
  
exit  
  
interface g0/2/2  
  
switchport mode access  
  
switchport access vlan 50  
  
no shutdown  
  
end
```

5.0 Problems and Solutions

During the project, I faced one big problem with the hardware.

The Problem:

The router did not have enough ports for 5 departments. I added a module card, but the new ports were "Switch Ports," not "Router Ports." This means I could not just type an IP address onto them. When I tried, the router gave me an error.

The Solution:

I learned that I had to use VLANs (Virtual LANs) for the extra ports.

1. I created VLAN 30, 40, and 50.
2. I assigned the IPs to the "Virtual Interface" (SVI) instead of the physical port.
3. I connected the cables to the module ports and told the router to put that traffic into the correct VLAN.

This fixed the issue, and now Sales, Finance, and HR work perfectly.

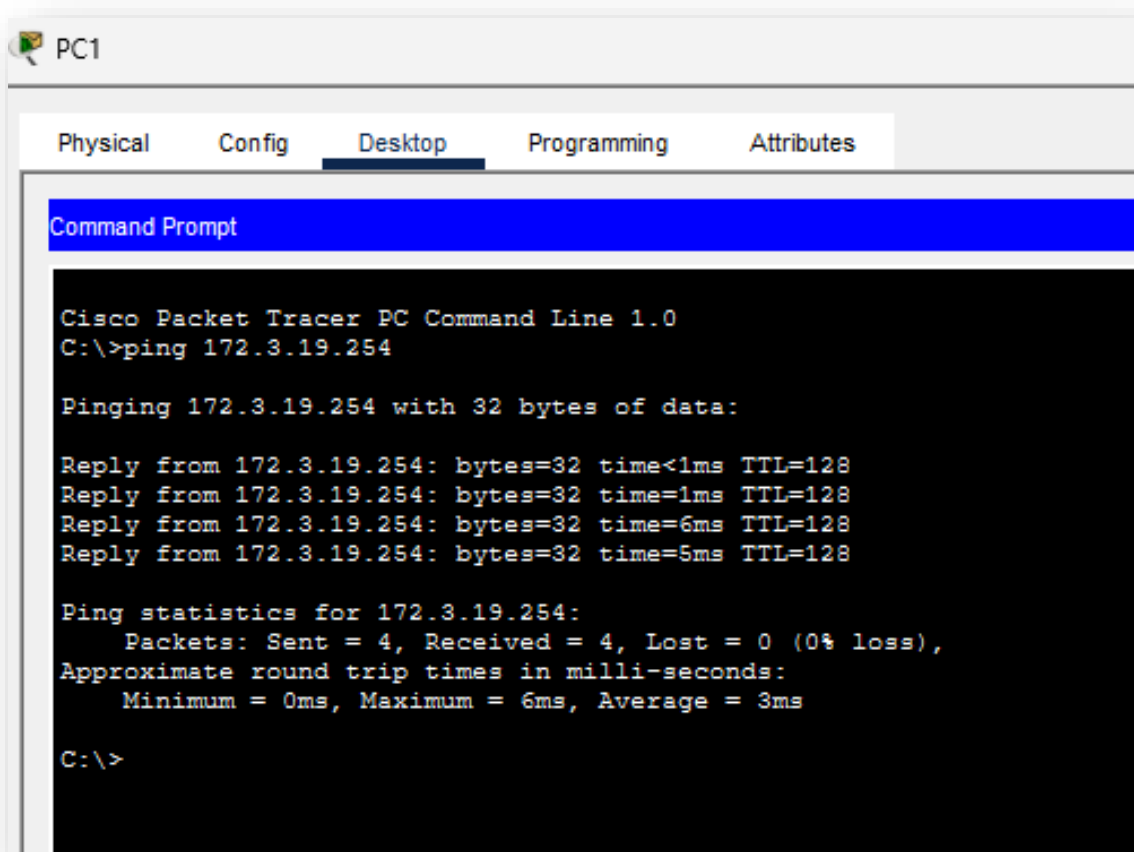
6.0 Verification (Proof)

To prove the network works, I did a "Ping Test."

6.1 The Ping Result

I sent a ping from the Engineering PC to the Finance PC.

- **Source:** 172.3.19.254
- **Destination:** 172.3.22.126
- **Result:** Successful (Reply received).



The screenshot shows a PC1 window with tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt window. The Command Prompt shows the output of a ping command from C:\> to 172.3.19.254. The output indicates that the ping was successful, with 4 packets sent and received, 0% loss, and round trip times ranging from 0ms to 6ms.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.3.19.254

Pinging 172.3.19.254 with 32 bytes of data:

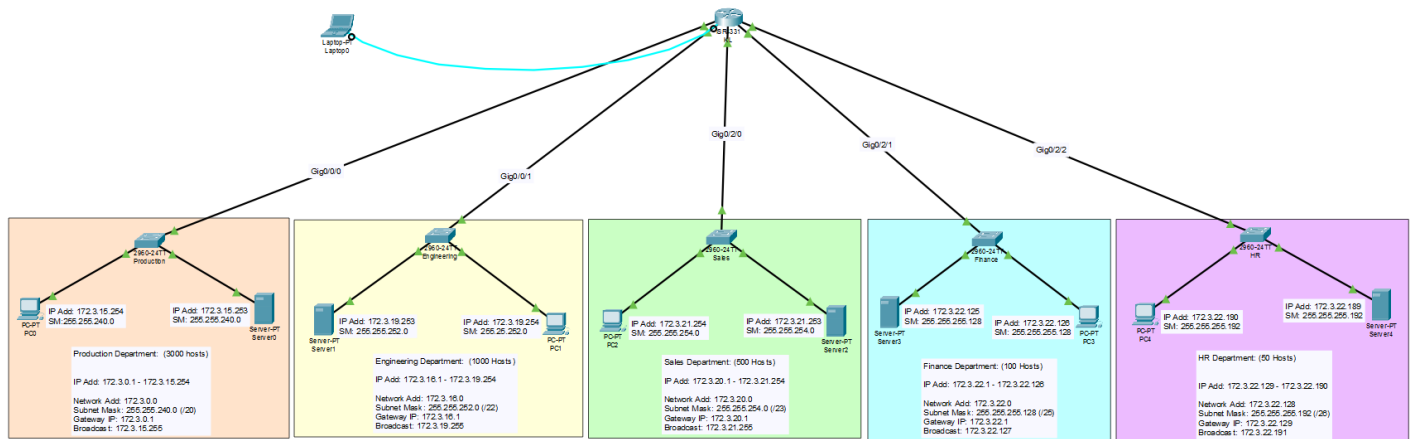
Reply from 172.3.19.254: bytes=32 time<1ms TTL=128
Reply from 172.3.19.254: bytes=32 time=1ms TTL=128
Reply from 172.3.19.254: bytes=32 time=6ms TTL=128
Reply from 172.3.19.254: bytes=32 time=5ms TTL=128

Ping statistics for 172.3.19.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 3ms

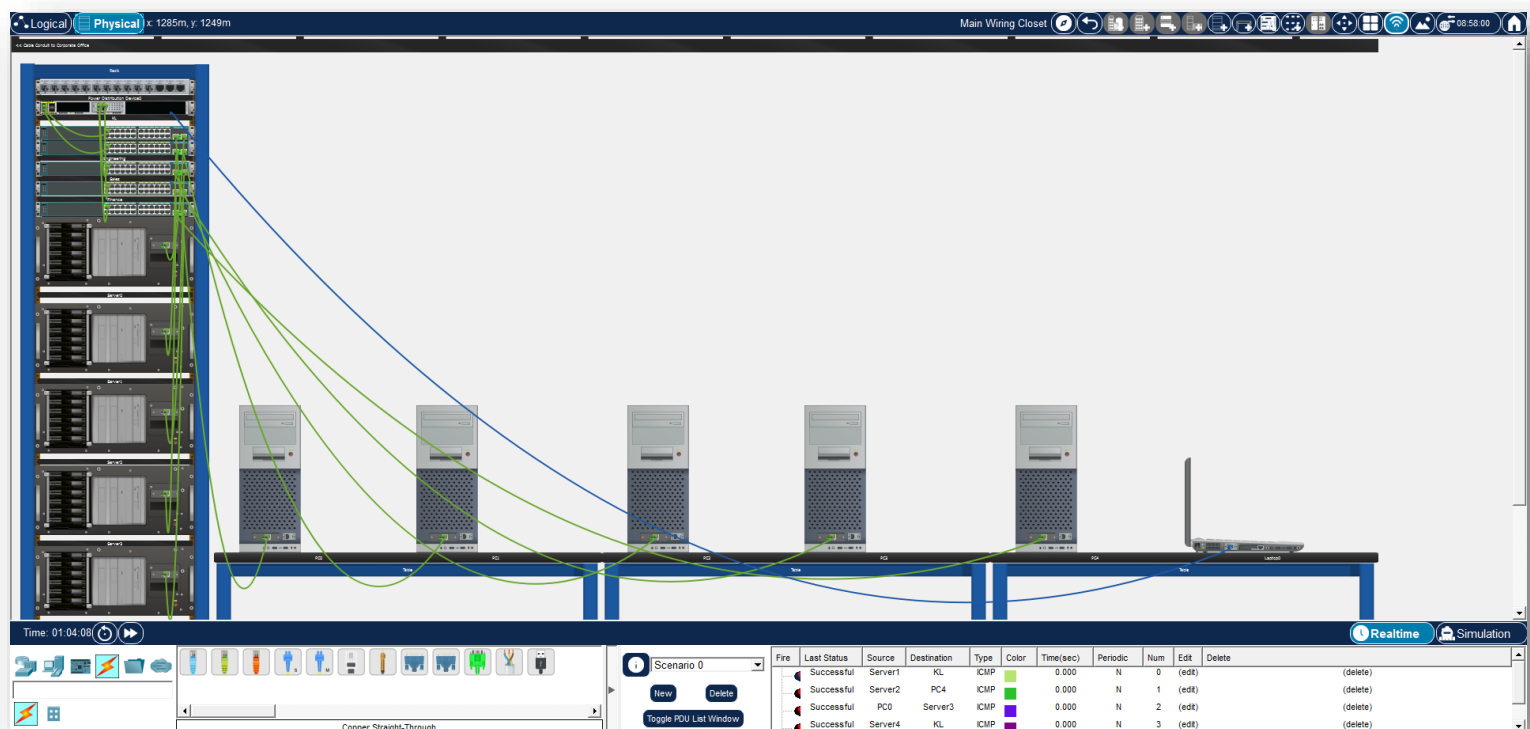
C:\>
```

6.2 The Topology Diagram

Here is the picture of the final network. All the lights are green, which means the cables are connected correctly.



Realtime Simulation										
Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	Server1	KL	ICMP	Green	0.000	N	0	(edit)	(delete)
	Successful	Server2	PC4	ICMP	Green	0.000	N	1	(edit)	(delete)
	Successful	PC0	Server3	ICMP	Purple	0.000	N	2	(edit)	(delete)
	Successful	Server4	KL	ICMP	Purple	0.000	N	3	(edit)	(delete)



7.0 Conclusion

In this assignment, I successfully designed a network for five departments. By using VLSM, I saved many IP addresses. I also learned how to mix normal router ports with a switch module using VLANs.

The final tests show that all computers can talk to each other. The network is ready to use.